

his memoir that had American diplomats not intervened, the conflagration risked spiraling and leading to nuclear use. Several months after the attack Indian Defense Minister Rajnath Singh, standing at the site of India’s second nuclear tests, announced that although India had previously “strictly adhered” to its NFU, “what happens in future depends on the circumstances” (Bagchi 2019). In May 2025, India and Pakistan engaged in a skirmish similar to that in 2019, during which India struck a Pakistani air base near the military headquarters responsible for overseeing its nuclear arsenal. India has also built out its intelligence gathering operations, improved the range of its ballistic missile system, and developed multiple targetable re-entry vehicles that would enable it to launch a counterforce attack against Pakistan that would disable its nuclear weapons facilities (Clary and Narang 2019).

Even as public officials have raised doubt about the legitimacy of India’s NFU, they have clung to the language of India being a “responsible” nuclear state. In the same speech where he questioned that India would abide by its NFU, Singh said that “India attaining the status of a responsible nuclear nation is a matter of national pride.” After public outcry over the BJP’s election manifesto, Modi called India’s NFU “a reflection of cultural inheritance” (Bushman 2014). The contradictory nature of these statements suggests that although elites may be disillusioned with the utility of India’s NFU, they still believe that it is a useful signaling device to either adversaries or a domestic public skeptical of an aggressive nuclear posture.

4 Research Design

To test whether identity-based arguments and knowledge of India’s NFU reduce support for nuclear use, I conduct a conjoint survey experiment modeled off of those created by Sagan and Valentino. In the experiment, respondents read a scenario describing a hypothetical Indian counterforce attack against Pakistan. The situation described is consistent with India’s Cold

Start doctrine and inspired by what security scholars consider to be the most likely scenario in which India would first use nuclear weapons (Kapur 2008; Clary and Narang 2019). Implemented after the Kargil War, Cold Start is military doctrine enabling India to quickly mobilize along the border with Pakistan following a state-sponsored terrorist attack. Within days of mobilization, India can push across the borders with Punjab and Kashmir either to claim disputed territory or force a favorable compromise.

Although India publicly maintains a no first use policy, Christopher Clary and Vipin Narang argue that during such a ground war, India may contemplate launching a nuclear attack. Because India's military is conventionally superior to Pakistan's, the latter could consider using a small-yield tactical nuclear weapon on the battlefield to stop an Indian incursion. However, if Pakistan is incentivized to use nuclear weapons, India is encouraged to take first mover advantage and do the same. Fearing an eventual strike against one of its cities, Clary and Narang believe that India could launch a counterforce attack, where it tries to eliminate Pakistan's strategic nuclear weapons. With the risk of a large-scale Pakistani attack marginalized, India could then conduct the ground war on its own terms.

A priori, it is difficult to know how successful such an attack would actually be, given uncertainty over where Pakistan's nuclear weapons facilities are, how effective its air defenses would be, and the ordinary risks entailed by any large-scale military operation. In the article, I note that it is unclear whether all of Pakistan's strategic nuclear weapons have been destroyed, but that Indian government officials speaking on the condition of anonymity believe that there is a low chance of retaliation. I do so as to create a more difficult test for the norm against nuclear use while not misleading respondents into believing that a counterforce attack would be guaranteed to work. In a similar vein, although it is difficult to gauge how many civilians would die in a counterforce attack, I note that the World Health Organization warns that it could reach into the hundreds of thousands. Again, I do so as to not mislead respondents into believing that a counterforce attack could be committed with little harm to civilians, even if the extent of that harm is ambiguous.

The story is structured like a mock news article so that key pieces of information can be highlighted in the headline, subtitle, and pull quotes.⁶ Specifically, I emphasize information that could influence people’s understanding of the strategic utility of the attack and whether it was morally justifiable—including the target of the attack, that it was launched during an ongoing ground war, the number of civilian casualties, and the risk of retaliation.⁷ In the article, I avoid primes that would prematurely invoke either people’s partisan identities or different conceptions of state identity. Without naming the prime minister, I refer nebulously to one with masculine pronouns. By virtue of Narendra Modi currently holding office, people may imagine that his government launched the attack. However, in the body of the text, I do not actively try to stimulate this thought. Across treatment and control groups, the article is identical, except that people assigned to learn about India’s NFU read an additional sentence saying that “the nuclear attack violates long-standing Indian policy that it would never use nuclear weapons unless first attacked with weapons of mass destruction.” A pull quote in the middle of the article draws further attention to this fact.⁸

After reading the vignette, respondents are assigned to either reading or not reading three arguments against nuclear use.⁹ The arguments are written to invoke India’s personal, positive identity and its negative, social identity vis-a-vis nuclear use. The first two directly mention Gandhi, Nehru, and India’s legacy of non-violence. I invoke Gandhism, specifically, because his deontological ethics are evocative of the nuclear taboo. In addition to calling on these historical figures, the second and third juxtapose India with other states, saying that the use of nuclear weapons violates India’s unique history as an advocate for global disarmament. The second explicitly mentions the U.S. and U.K.,. Mentioning the only

⁶All experimental materials can be found in Appendix B.

⁷Note that in the article, the prime minister himself says that India was facing no imminent threat of nuclear attack from Pakistan. In reality, it is implausible that a political leader would make such a confession, given the substantial domestic and international benefits of misrepresenting their motivations. That said, telling respondents that a nuclear attack was imminent would be an unreasonable test of the nuclear taboo because it would trigger people’s instinct for self-defense against immediate threats.

⁸Note that to ensure no respondent mistakenly believes that the attack described is real, they are told both before reading the vignette and upon exiting Qualtrics that the scenario is hypothetical.

⁹This structure creates a 2x2 factorial experiment. Treatment conditions were block randomized on the basis of party.

state to have used nuclear weapons and the one that colonized the subcontinent may serve to bolster nationalist claims that India has a unique moral stature. In both the first and second argument, there are appeals in the first-person plural. Evoking a conception of “we” is important because norms are inter-subjective. It is not sufficient for one person to hold a reservation about a given course of action, but it must be held among a group of people. The third argument does not include specific mentions of Indian historical figures or “we-ness,” but refers broadly to India as having been a responsible state and upholding moral leadership.

To measure how strongly the public supports nuclear weapons use, I include six different survey questions. In keeping with existing survey experiments, I ask respondents how strongly they approve of the nuclear attack as described in the vignette and how strongly they would prefer it to an alternative (here, continuing the ground war with Pakistan). Approval is the most direct way to measure whether someone supports nuclear use. Approval is also relevant to elected politicians fearful of domestic backlash and keen on limiting public opposition to their actions. Asking people whether they would prefer nuclear use to another military action is valuable because it reveals how strongly they have internalized just war theory. Whereas the nuclear attack described could not be defended under just war theory because of its failure to discriminate between combatants and civilians, as well as its lack of proportionality, the ground war would be defensible depending on the style and extent of combat, which is not described in the vignette.

In addition to asking for people’s approval and preferences, I ask two questions regarding what material action they would take to punish or not punish the sitting government. Specifically, would they be more or less likely to vote for the ruling party, and how likely would they be to protest nuclear use? Similar questions have yet to be asked in extant survey experiments, but offer a more direct test of whether politicians would face reprimand for nuclear use than simply asking for public approval. Though approval is a useful measure of overall public support for a military action, it is only relevant to an election-seeking

politician who believes that it will affect how people vote. If a given policy decision is widely unpopular but low salience, then it will have little effect on voting patterns. By contrast, if the reaction to an action is polarized but high salience, then it could either incentivize or disincentivize a politician from supporting nuclear use, depending on the distribution of voters who support it.

Lastly, I include measures of how moral respondents believe the attack to be. One of the questions asks directly whether respondents believe that the nuclear attack was ethical. Another asks whether they think it violates *ahimsa*. These questions are useful insofar as they may reveal consequentialist reasoning. If a respondent believes that an attack is not moral but approves of it nonetheless, then that could imply they are willing to justify the attack based on the military and strategic benefits that it produces.

All outcome variables are measured on seven point Likert scales, with respondents being given the option to select “don’t know.” If I were to find evidence to substantiate my hypotheses, I could observe two different, potentially compatible, patterns in people’s responses. On one hand, support for nuclear use could decline across the seven point Likert scale. In this scenario, people’s support for nuclear use changes with little difference in the number of survey takers reporting that they don’t know how to respond. In another scenario, the incidence of people reporting that they don’t know how to respond could increase while the number of people supporting the attack decreases. In this case, the treatments convert people into fence-sitters. Instead of being in favor of military action, they are unsure of how to respond. To account for this possibility, I run all of my analyses twice. In one case, I exclude all NA observations; in the other, I impute values for people who say “don’t know,” placing them in the middle of the Likert scale for each of the my outcomes.

To gauge the extent of public support for nuclear use, I create an index using inverse covariance weighting that represents respondents’ cumulative opposition to nuclear use (Anderson 2008). To create this index, I use the public’s approval of the attack, whether it would affect how they vote, and whether they would publicly protest it. I group these fac-

tors together because they are indicative of how strongly a politician would be sanctioned for nuclear use. Creating such an index is useful to my study both practically and theoretically and is an innovation on previous scholarship evaluating the taboo. ICW is useful practically because it enhances statistical power and reduces the number of tests that I run, minimizing the multiple comparison problem. Theoretically, ICW is useful because an election-seeking politician will weigh several aspects of public opposition to a nuclear attack when deciding whether to conduct one—not strictly, for example, public disapproval of the attack or whether people would protest it. As a result, though ICW does not provide an easily interpretable outcome variable, the index is holistic and provides insights into the different possible sources of opposition to nuclear use.

Previous experimental research on the taboo has not taken this approach, preferring to measure average treatment effects by looking at how treatments alter overall approval of nuclear use or people’s preference for an alternative. This existing approach, I believe, both understates the degree of public opposition to norm-breaking behavior and mischaracterizes Tannenwald’s initial theorizing on the taboo. Should it be the case, as Sagan and Valentino (2017) found, that there is no meaningful difference between the percentage of people who would approve of a nuclear attack that killed one hundred thousand versus two million Iranian civilians, then how does the increased number of casualties affect the strength of opposition among the minority? Would it cause more people to protest, vote against the government, and otherwise politically punish the U.S. president for launching such a destructive attack? Irrespective of whether a majority of people would approve of the attack, if there is a high willingness among a meaningful subset of the total population to punish the sitting government, then domestic political opposition could be a constraint on norm-breaking behavior. An index-based approach is in-keeping with Tannenwald’s initial theorizing on the taboo because she locates the strength of public opposition to nuclear use not only among the sheer number of people who would disapprove of it but also in the grassroots anti-nuclear movements that helped to animate people’s moral revulsion. Based on these benefits of an

index-based approach, I estimate the following OLS model as the primary means to evaluate hypotheses two through four:

$$Y_i = \alpha + \beta_1 Identity_i + \beta_2 NFU_i + \beta_3 (Identity_i * NFU_i) + \beta_{party} + \delta' \mathbf{X}_i + \epsilon_i, \quad (1)$$

where i indexes respondents, Y_i represents composite attitudes toward nuclear use, β_1 is the fixed effect of reading identity-based arguments, β_2 is the fixed effect of telling respondents about India's NFU, β_3 is the fixed effect of the interaction between both treatments, β_{party} is a block fixed effect for party, and X_i is a vector of LASSO-selected individual-level covariates, including gender, religion, age, education, and whether someone is from a state that borders Pakistan. Y_i is constructed such that the higher values reflect stronger opposition to nuclear use.

Although the above model is useful for gauging whether my treatments have an effect on public opposition to nuclear use in the manner that I expect, it has several limitations. Most notably, it cannot be used to evaluate my first hypothesis. To determine whether the public has a categorical aversion to nuclear use, we must look at the sheer magnitude of support in the control group. I do so by collapsing responses to the six measures of my outcome into three categories: support, indifference, and opposition. I reduce my outcomes into three categories instead of two because indifference does not complement arguments either that there is a taboo against nuclear use or that people are particularly receptive to it. As a result, to generate a descriptively accurate assessment of public opposition to nuclear use, I do not group indifference with either approval or disapproval of the attack.

Second, even though ICW is a useful way to create a composite score for support of nuclear use, it weights different measures in a manner that's theoretically arbitrary. To be sure, factor analysis or another means of creating a composite score for public opposition to nuclear use would struggle from the same problem. Because different leaders would not

weigh different sources of public opposition to nuclear use in the same manner, any particular weighting system that researchers impose would not universally apply. That is to say nothing of the fact that because ICW and factor analysis do not produce easily interpretable results, they would not be useful for political elites deciding whether to use nuclear weapons.

I account for these problems in two ways. First, to test the robustness of my findings, I implement principal component analysis (PCA). Although this approach does not resolve issues regarding the interpretability of my outcome variable, it will show whether my treatments are still statistically significant using a different weighting scheme, reducing concerns that any statistically significant finding is attributable to one configuration of my composite outcome. Second, I estimate a series of linear probability models, in which I dichotomize every outcome variable such that indifference to and disapproval of the attack are grouped together. Regressing each outcome on my variables is useful because ICW and PCA are “black boxes.” That is to say, it is not clear which measures of my outcome are being up- or down-weighted. By breaking apart respondents’ answers across each of my outcomes, I evaluate how my treatments affect each type of opposition distinct from one another. Although this approach raises the problem of multiple comparisons, it reduces concerns regarding the theoretical arbitrariness of using any given weighting scheme. As a result, by evaluating the effects of my treatments on both indices and those indices’ constituent parts, I am able to use one approach to account for the deficiencies of the other. In the appendix, I include further robustness tests.

To conduct my experiment, I use Prolific to recruit four hundred participants to fill out a survey on Qualtrics. I select Prolific primarily for its ease of use compared to other survey platforms.¹⁰ Being cheap to recruit respondents, Prolific enables me to gain enough statistical power to adequately test the hypotheses that I proposed above. That being said, online surveys in India are rarely representative of the population. In India, online samples

¹⁰I initially planned on recruiting participants via Amazon MTurk, which would have enabled me to acquire a larger sample. However, due to last minute difficulties with MTurk’s customer service team, I opted for Prolific.

are disproportionately young, male, upper-caste, and wealthy (Boas, Christenson and Glick 2020). In an effort to account for some demographic differences, I weight my sample by gender and age. That said, by virtue of having such an unrepresentative sample, there is only so much that balancing can correct for. Furthermore, because my survey is conducted in English, it cannot reasonably be construed as representative even after weighting. Because English is most widely spoken among people who are well educated, my findings will likely only apply narrowly to upper-class, elite opinion.

Having an upper-class sample has both benefits and drawbacks. Assuming that people who are better educated are more norm sensitive, an elite sample may constitute a most likely case to observe a widespread normative aversion to nuclear use. Should a taboo not exist in my sample, then it may be less likely to exist more broadly. Likewise, given how well educated English-speaking people are on balance, people may know more about Indian history and foreign policy than the median voter, meaning that identity-based appeals could be more salient than they otherwise would be. In spite of this potential benefit of an elite sample, it is nonetheless not ideal. If we are concerned with the absolute magnitude of public opposition to nuclear use—and whether that opposition would manifest in people protesting or voting against the government—then it is important that we know how people beyond the upper crust think. However, based on the nature of online surveys in India and this survey being conducted in English, we cannot confidently draw conclusions about Indian public opinion writ large.

5 Results

The discussion of my results proceeds in six parts. First, I consider how strongly the anti-nuclear norm is rooted in India and whether it plausibly constitutes a taboo. Second, I evaluate the effects of my treatments on public attitudes toward the counterforce attack using ICW. Third, I break down the effects of my treatments on each of the components